Problem Set 5

Keep your rendered `.pdf` to no more than 5 pages. Only provide code in the rendered `.pdf` when it is specifically requested.

1. Type `data(mtcars)` to load in the `mtcars` dataset.

(a) How many observations (rows) are in this data and how many variables (columns)?

(b) Create a table using `kable` that summarizes the minimum, mean, maximum, and standard deviation for each of the following variables: `mpg`, `hp`, and `wt`. Give the columns descriptive names e.g. "Miles per Gallon" for `mpg`. You'll want to create a table with 4 rows and 3 columns.

(c) Make a plot of the car weights against their miles per gallon. Make sure that your plot is clearly annotated and readable.

2. We can download data on fruit prices from the USDA: https://www.ers.usda.gov/data-products/fruit-and-vegetable-prices. Download the file linked below "ALL FRUITS – Average prices (CSV format)."

(a) What character separates distinct fields in this file?

(b) Print the code you use to read in the unaltered file using `read.csv` to the rendered `.pdf`.

(c) How many rows does this dataset contain? Print the code you use determine this to the rendered `.pdf`.

(d) How many columns does this contain? Print the code you use determine this to the rendered `.pdf`.

(e) What variables are contained in this dataset? Print the code you use determine this to the rendered `.pdf`.

(f) What fruit has the highest retail price? Print the code you use determine this to the rendered `.pdf`.

(g) What fruit has the highest price, normalized for quantity (this is the `AveragePricePerCupEquivalent` variable)? Print the code you use determine this to the rendered `.pdf`.

(h) How many fresh fruits are described in this dataset? Print the code you use determine this to the rendered `.pdf`.

(i) What happens if you remove all quotes from the file and then read in the file using `read.csv`. Explain in at most one sentence.

3. On my teaching page, you can find a dataset summarizing some Massachusetts employment statistics called `CESReport.csv`. It was downloaded from here: https://lmi.dua.eol.mass.gov/LMI/CurrentEmploymentStatistics

(a) Read this data into `R` using `read.csv`. You want to obtain a data frame with 172 observations and 15 variables. Print the code you use determine this to the rendered `.pdf`.

(b) What are the modes of the variables in this data?

(c) What happens if you apply `as.numeric` to the `Year` variable? Explain in at most sentence.

(d) What happens if you apply `as.numeric` to the `January` variable? Explain in at most sentence.

4. For this problem, we'll keep working with the baseball database that we have used in class.

(a) Create a table with 1 rows and 3 columns, where each column corresponds to one of the following datasets, "Salaries", "Master", "Batting", and each element describes the number of observations in the corresponding dataset using `kable`. Make sure that the table is self contained and readable. Print the code you use obtain the number of observations per dataset to the rendered `.pdf`.

(b) Using `dbGetQuery`, read in all observations for all variables in the "Batting" data. Print the code you use to the rendered `.pdf`.

(c) Using `dbGetQuery`, read in all observations for the `playerID`, `yearID`, and `HR` variables in the "Batting" data. Print the code you use to the rendered `.pdf`.

(d) Now we are going to start conducting some timing comparisons. Recall that in `arcovariance.R` file on my teaching page we used the `Sys.time()` function to time different operations in `R`. We will use it again here. Record the time it takes to perform the tasks in (b) and (c). Print the code you use to the rendered `.pdf`.

(e) Provide your results from (d) and explain why they do or do not make sense, in at most one sentence.

(f) Using `dbGetQuery`, read in all observations for the `playerID`, `yearID`, and `HR` variables in the "Batting" data. **Then** subset the data to just observations with year 2000 or later. Print the code you use to the rendered `.pdf`.

(g) Using `dbGetQuery`, read in observations during and after 2000 for the `playerID`, `yearID`, and `HR` variables in the "Batting" data. Print the code you use to the rendered `.pdf`.

(h) Record the time it takes to perform the tasks in (f) and (g). Print the code you use to the rendered `.pdf`.

(i) Provide your results from (h) and explain why they do or do not make sense, in at most one sentence.